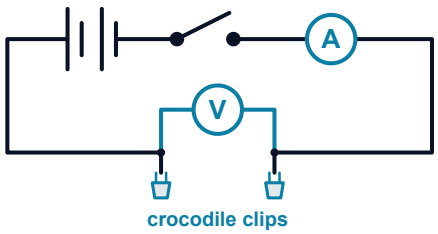
DO NOW · REQUIRED PRACTICAL

# Resistance and the length of a wire

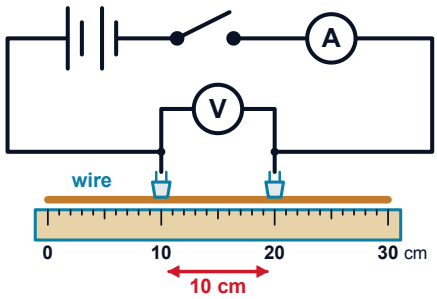
**Electricity**
Paper 1 · Combined and Separate
6 steps

A student investigates how the length of a wire affects its resistance.

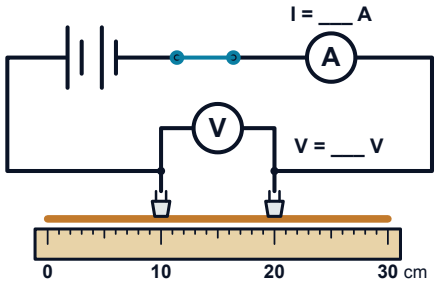
**1** **Set up a circuit with an ammeter in series, ending in two crocodile clips. Connect a voltmeter across the clips.**



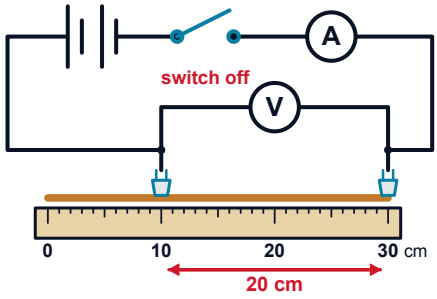

**2** **Attach the crocodile clips 10 cm apart on the wire, so the voltmeter is across this length of wire.**




**3** **Switch on and record the current and the potential difference.**




**4** **Switch off to let the wire cool, then move the clips to 20 cm apart.**



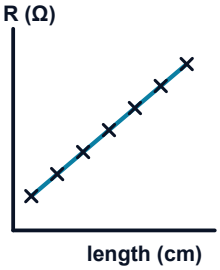

**5** **Repeat the readings for a range of lengths.**

| length (cm) | I (A) | V (V) |
|---|---|---|
| 5 | | |
| 10 | | |
| 15 | | |
| 20 | | |
| 25 | | |
| 30 | | |

**6** **Calculate the resistance for each length and plot resistance against length.**

$$R = V \div I$$

(for each length)




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